

NETWORK INTRUSION DETECTION AGENT

Decides whether to allow / block / flag network connections in real time, using a trained classifier's confidence plus recent host traffic history.



ENVIRONMENT

Property	Classification	Why
Observable	Partially Observable	Only sees connection features, not attacker intent
Deterministic vs Stochastic	Stochastic	Same features can mean different real outcomes
Episodic vs Sequential	Sequential	Past connections to a host affect current decision
Static vs Dynamic	Dynamic	Attacker is actively sending new connections while the agent is deliberating
Discrete vs Continuous	Discrete	Action space has exactly 3 options: allow, block, flag
Single vs Multi-agent	Multi-agent	Human reviewer acts on flagged connections, and attacker is actively trying to evade, both are agents

THE DATA

NSL-KDD dataset (Kaggle), network connection records

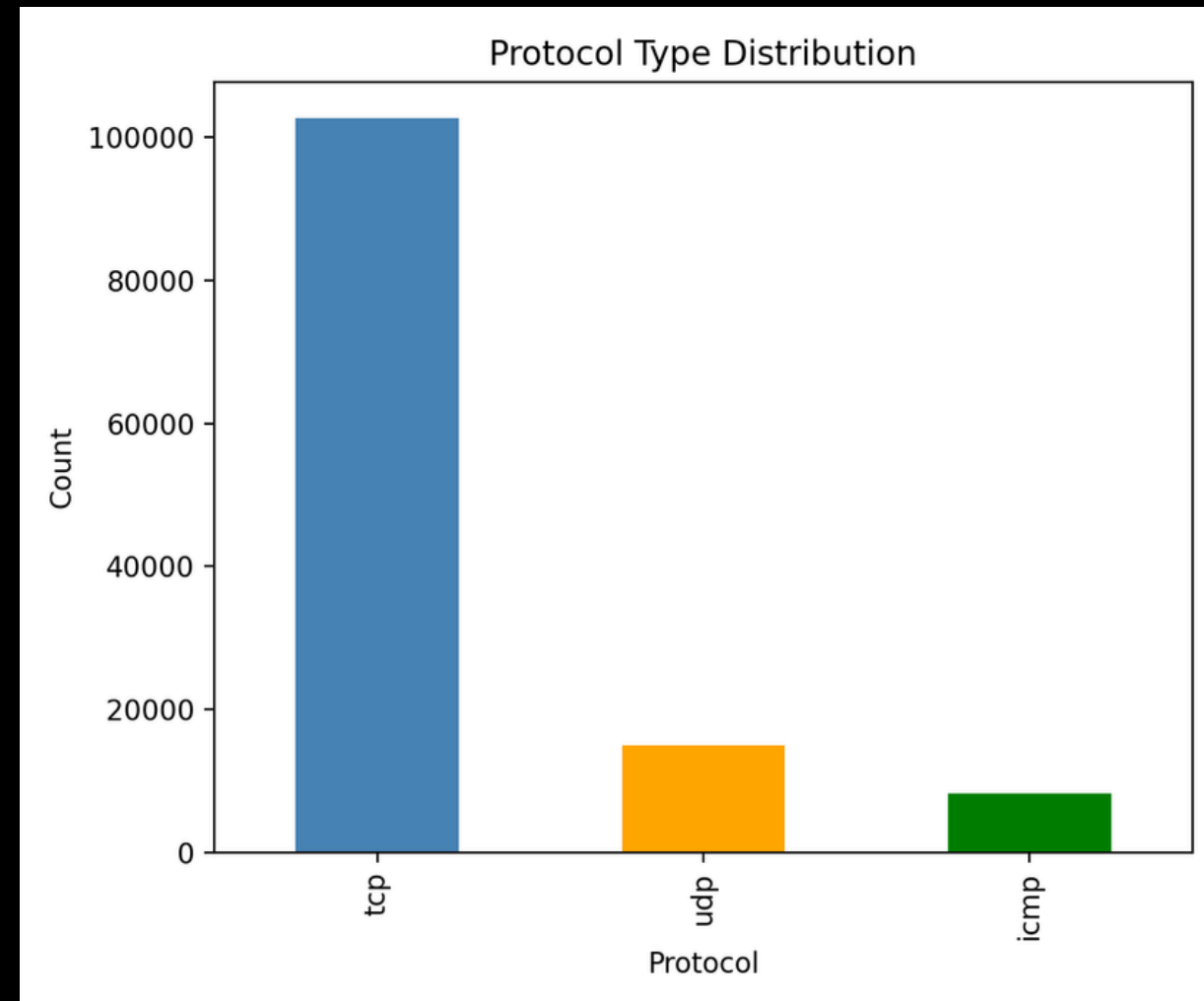
125,973 training rows. 22,544 test rows.
41 features: duration, protocol, byte counts, error rates...

Labels: normal or 1 of 22+ attack types

Data issues:

Class imbalance: u2r attacks have only about 40 examples vs about 13,000 normal connections

Test set has unseen attack types not in training, which tests real generalization



Distribution of connections by protocol type (tcp/udp/icmp)

duration	protocol_type	service	flag	src_bytes	dst_bytes	label
0	tcp	ftp_data	SF	491	0	normal
0	udp	other	SF	146	0	normal
0	tcp	private	S0	0	0	neptune
0	tcp	http	SF	232	8153	normal
0	tcp	http	SF	199	420	normal

Sample of raw connection records from the dataset

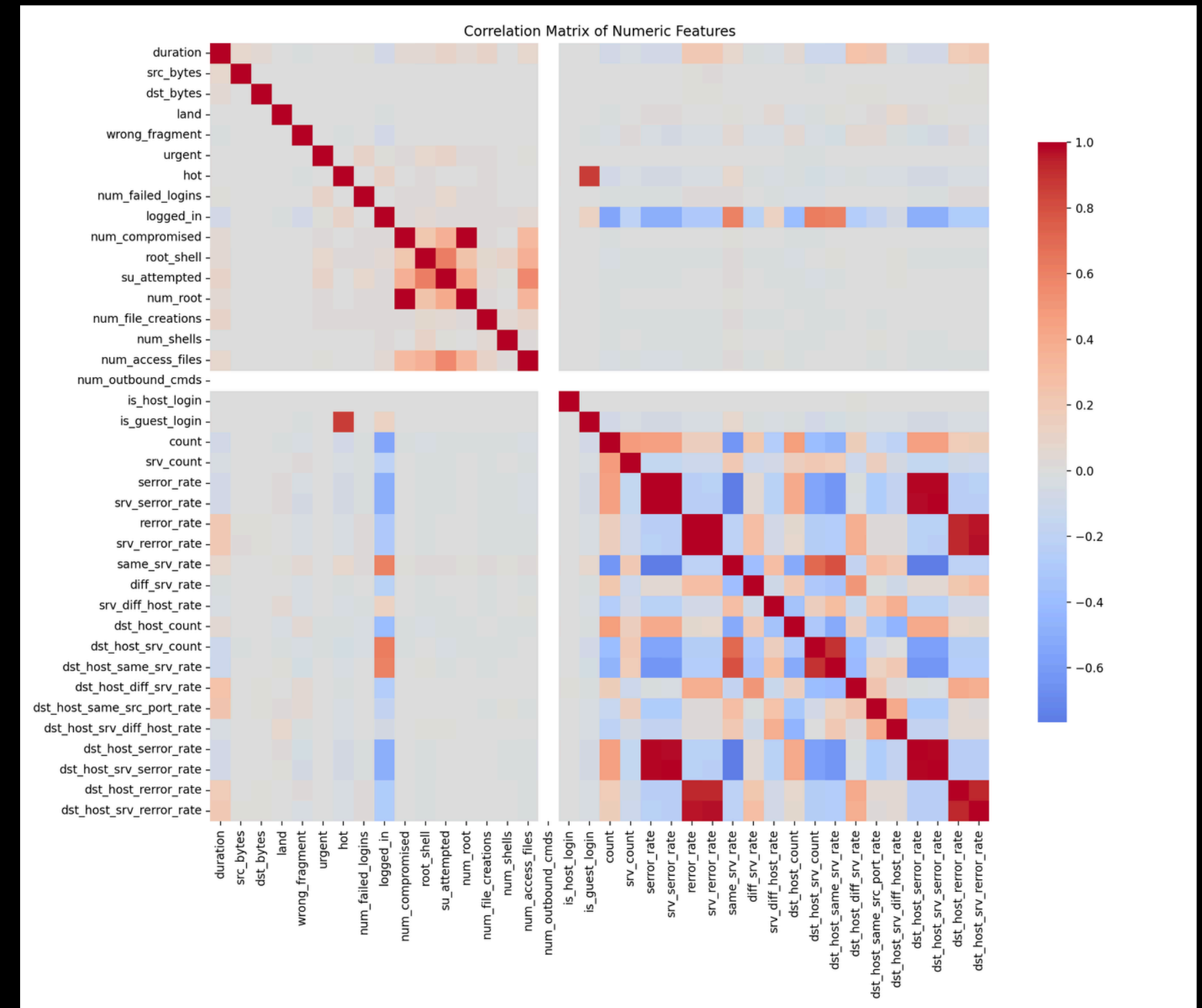
DATA ANALYSIS

Correlation matrix across 38 numeric features

Privilege escalation features cluster together (root_shell, su_attempted, num_root)

Error rate features correlate strongly within their own group

Confirms structure in the data before modeling



Correlation matrix across 38 numeric features, showing which measurements move together

MODEL

Supervised Learning, Decision Tree
(max_depth=8)

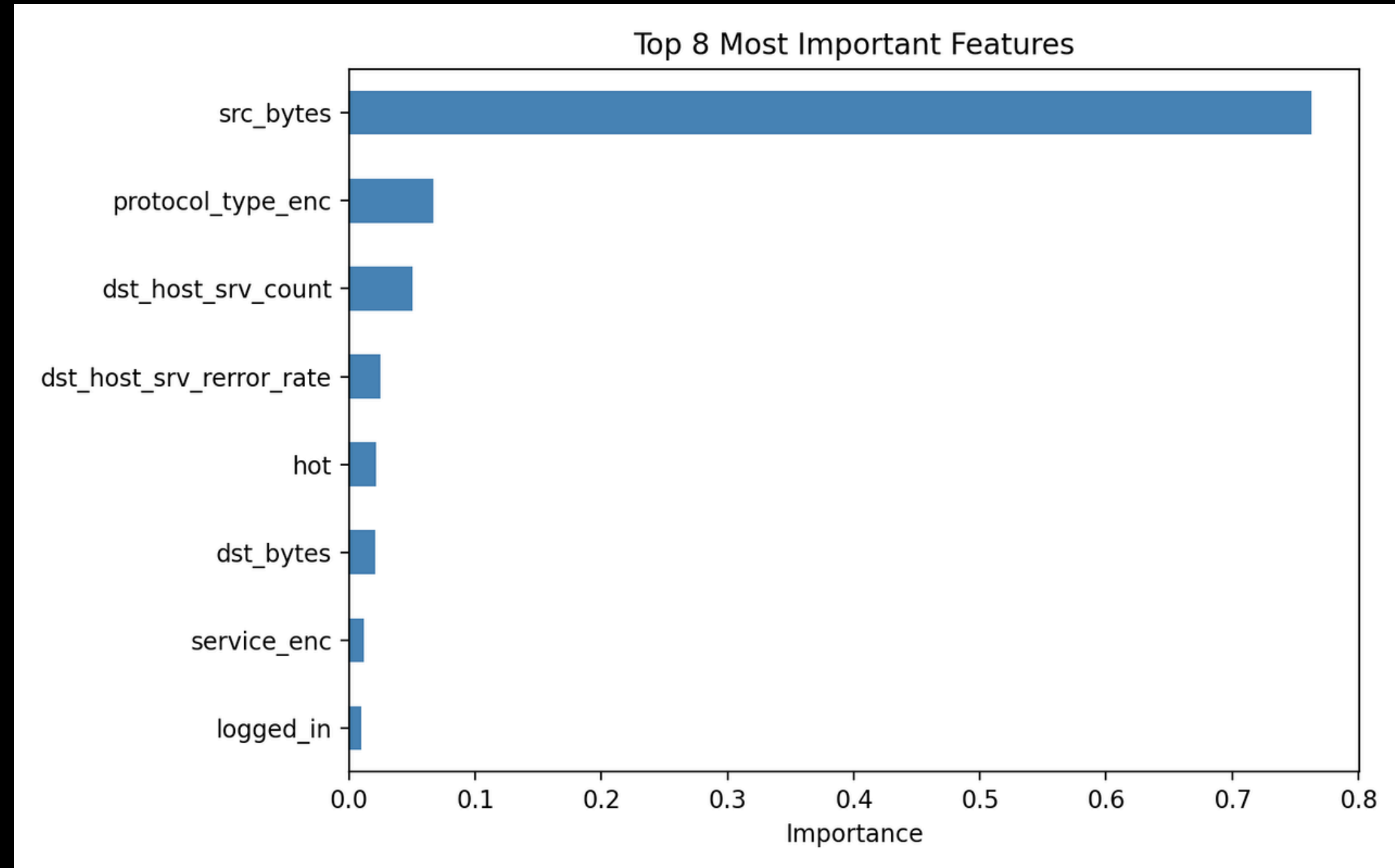
Two versions trained: binary (normal/attack)
and multi-class (dos/probe/r2l/u2r)

Model to Agent pipeline:

Prediction + confidence, then a decision
policy (threshold 0.90), then allow, block, or
flag

Sequential rule:

High recent host activity plus high error rate
escalates a connection to "flag"



Top features the Decision Tree relies on most when
classifying a connection

RESULTS

Binary (real test set): 77% accuracy

Multi-class (real test set): 83.8% accuracy

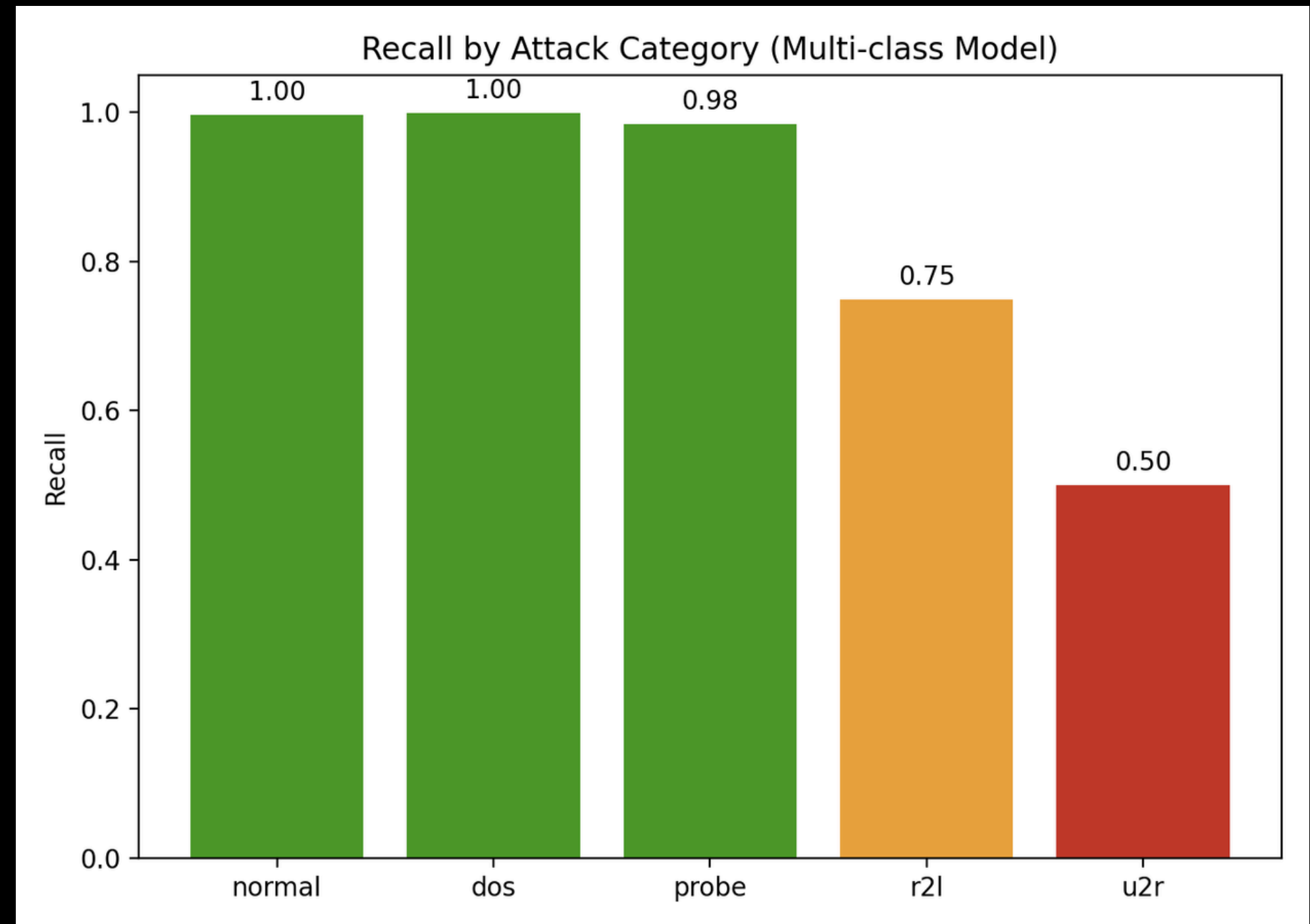
Multi-class (random split): 99.4% accuracy

The big number: 15.6 point gap

99% is inflated. Real world performance on unseen attacks is 84%.

Rare attacks (r2l, u2r) are almost never caught, recall around 0.01 to 0.11

Improve with class balancing and ensemble models



Recall by attack category, showing the model misses rare attack types far more often

AGENT IMPLICATIONS

What the model gives the agent:
a prediction + confidence score, not a final answer

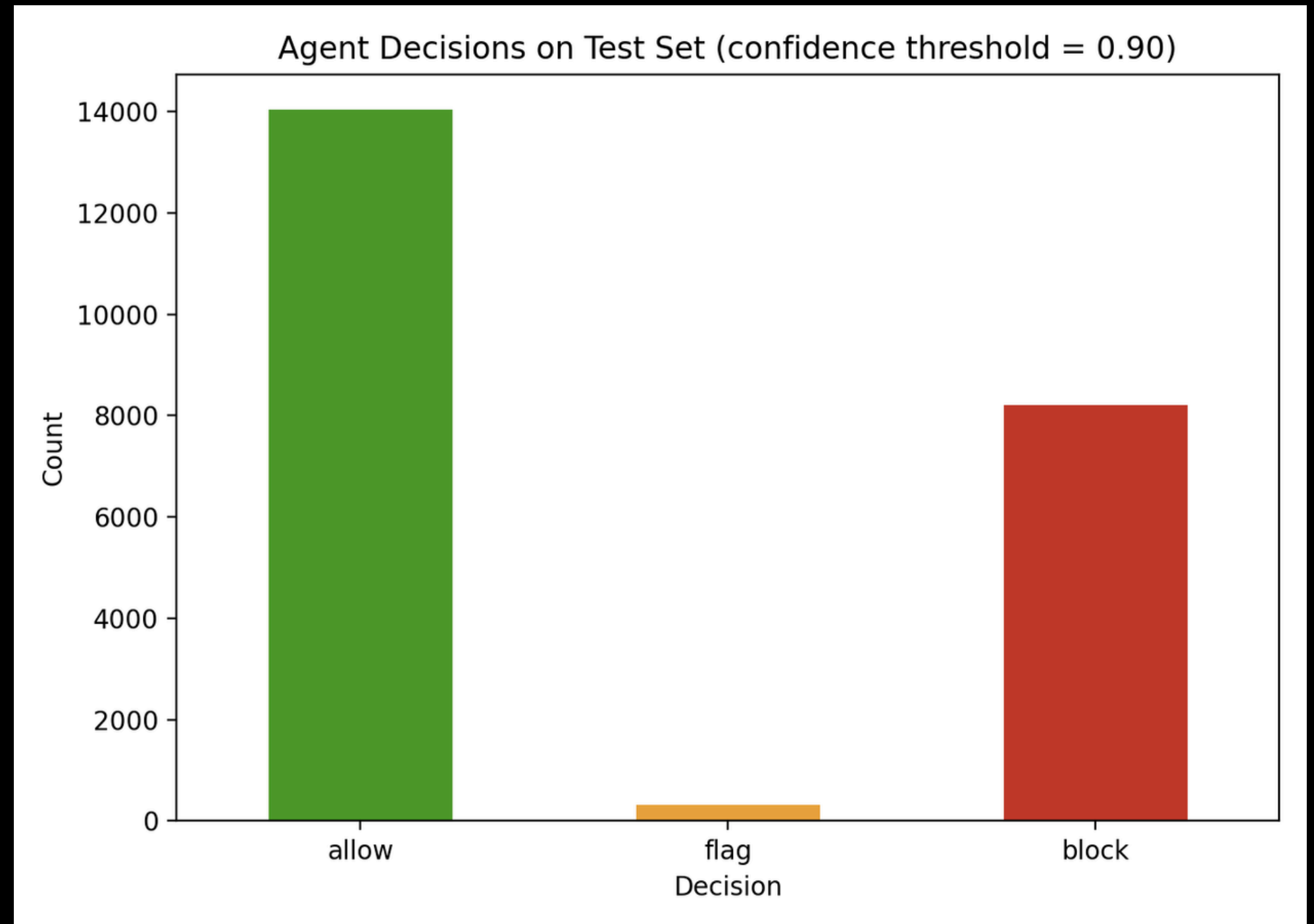
What it enables:

- Auto-block high-confidence attacks

- Flag low-confidence cases for human review

- Escalate using host history (307 connections re-flagged, 282 were real attacks)

Limitation: model alone isn't safe to fully automate, it misses 38% of attacks, almost blind to rare types. Needs a human-in-the-loop fallback.



Agent's decision breakdown at a 90% confidence threshold (allow / flag / block)